IN THE HIGH COURT OF JUSTICE
BUSINESS AND PROPERTY COURTS OF ENGLAND AND WALES
TECHNOLOGY AND CONSTRUCTION COURT (KBD)

Claim No. HT-2025-000612

B E T W E E N:

HALSTEAD AEROSPACE LIMITED

Claimant

- and -

FERNCLIFFE ALLOYS LIMITED

Defendant

EXPERT REPORT OF DR AILEEN FROST

In the field of metallurgy and materials engineering — on behalf of the Claimant

8 November 2024

1. Introduction and instructions

I am Dr Aileen Frost, CEng, FIMMM, a Chartered Engineer and consultant metallurgist with Argent Metallurgical Services. I hold a PhD in physical metallurgy and have 22 years' experience in titanium alloy metallurgy and aerospace materials failure analysis. I have been instructed by Whitfield Carter LLP on behalf of the Claimant to examine the titanium bar drawn from heats FA-7733 and FA-7741 and to express an opinion on (i) whether the material conformed to AMS 4928 and specification HAS-MAT-114, and (ii) the most likely cause of any non-conformity.

I understand my overriding duty is to the Court and not to the party instructing or paying me. I have complied with that duty. This report sets out my independent opinion on the matters within my expertise.

2. Material examined

I examined retained test coupons and sections taken from bar of both heats, together with the mill certificates, the certificates of conformity, the Claimant's incoming inspection records, and non-conformance report QN-2024-0337. I carried out optical microscopy, scanning electron microscopy with EDS, microhardness traverses, and independent chemical analysis by inductively coupled plasma and inert-gas fusion (for oxygen).

3. Findings — chemistry

Element / heat	AMS 4928 limit	FA-7733 (measured)	FA-7741 (measured)	Result
Aluminium (Al)	5.50 – 6.75 %	6.18 %	5.31 %	FA-7741 below min
Vanadium (V)	3.50 – 4.50 %	4.02 %	3.88 %	Conforming
Iron (Fe)	≤ 0.30 %	0.21 %	0.24 %	Conforming
Oxygen (O)	≤ 0.20 %	0.226 %	0.241 %	Both above max
Hydrogen (H)	≤ 0.015 %	0.006 %	0.008 %	Conforming

The aluminium content of heat FA-7741 (5.31%) falls below the AMS 4928 minimum of 5.50%. The interstitial oxygen content of both heats exceeds the 0.20% maximum. Elevated oxygen raises strength but reduces ductility and fracture toughness and is significant for fatigue-critical aerospace parts. The

certified chemistry on the mill certificates did not reflect these measured values.

4. Findings — microstructure

Optical microscopy of sections from both heats revealed a continuous oxygen-enriched surface layer (alpha-case) of between 95 and 180 micrometres in depth, confirmed by a microhardness gradient of more than 120 HV between the case and the bulk. I also observed localised beta-flecks, indicative of segregation during ingot solidification. Alpha-case of this depth is not acceptable on machined fatigue-critical parts and is outside the acceptance criteria of HAS-MAT-114.

5. Cause

The beta-flecks are a casting/solidification feature and can only have originated in the production of the ingot, that is, before delivery. As to the alpha-case, the depth and the uniform distribution around the full bar circumference, including on as-delivered (un-machined) reference sections retained by the Claimant, are consistent with the case having been present on the bar as delivered and inconsistent with localised introduction during the Claimant's machining, which does not subject the bar to the sustained elevated temperatures required to grow alpha-case of this depth. The Claimant's storage and handling records disclose no elevated-temperature exposure.

6. Opinion

1. The bar drawn from heats FA-7733 and FA-7741 did not conform to AMS 4928 (oxygen on both heats; aluminium on FA-7741) or to the microstructural acceptance criteria of HAS-MAT-114 (alpha-case and beta-flecks).

2. On the balance of probabilities the non-conformities were present in the bar as delivered on 2 February 2024 and were not introduced by the Claimant's subsequent processing.

3. Material in this condition is not suitable for the manufacture of safety-critical machined wing-rib fittings, and parts machined from it could not properly be certified for flight.

STATEMENT OF TRUTH AND DECLARATION

I understand that my duty is to the Court and I have complied with it. I confirm that I have made clear which facts are within my own knowledge and which are not, and that the opinions expressed are my own. I believe that the facts I have stated are true and that the opinions I have expressed are correct.

Dr Aileen Frost, CEng FIMMM — Argent Metallurgical Services — 8 November 2024